Cognitive Status Classification Using OASIS-1 Clinical and MRI Measures

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***Abstract*—This study examines whether demographic, structural-MRI summary, and cognitive-score variables distinguish records with a recorded Clinical Dementia Rating (CDR) of zero from records with CDR greater than zero in a supplied OASIS-1 cross-sectional workbook. The workbook contains 436 rows and 12 attributes; 235 rows have observed CDR labels and 201 have missing CDR. The labelled subset contains 135 CDR-zero records and 100 records with CDR above zero. Logistic regression was evaluated using staged feature blocks, alongside a shallow decision tree, k-nearest neighbours (KNN), a soft-voting ensemble, and a majority-class baseline. A stratified 60/20/20 train-validation-test split was used, with operating thresholds selected on validation data by the F_2 score. On the 47-record held-out test set, the ensemble achieved the highest ROC-AUC (0.893), the decision tree achieved the highest F_2 (0.893), and KNN achieved the highest accuracy (0.766) and specificity (0.741). The findings are exploratory: the test set is small, CDR is missing for a substantial fraction of the workbook, and MMSE is measured contemporaneously with CDR. Therefore, the results describe classification of recorded status and do not establish future dementia prediction or clinical utility.**

***Index Terms*—healthcare analytics, OASIS-1, cognitive status, Clinical Dementia Rating, MRI, machine learning, classification**

I. Introduction

Dementia-related research increasingly uses machine learning (ML) to examine patterns in cognitive assessments and brain imaging. Reviews report a broad range of model families and input modalities, but also note variation in cohort selection, validation design, reporting quality, and generalizability [1]–[3], [5]. MRI-based studies often work with raw images and deep neural networks, whereas clinical datasets may also contain compact numerical summaries that can be analysed using conventional classifiers [4], [6], [8]. These settings should not be treated as interchangeable: a model trained on tabular MRI-derived measures is not an image-based diagnostic system.

This project uses the supplied OASIS-1 cross-sectional workbook. OASIS was introduced as an open-access collection of MRI data spanning multiple age and cognitive-status groups [19]. The present analysis is narrower than the original dataset description: it classifies recorded CDR groups from tabular demographic, MRI-summary, and cognitive variables. The research question is: *how does test-set discrimination change when MRI-derived measures and then MMSE are added to demographic features, and how do several conventional classifiers compare?*

The contribution is a reproducible comparison design rather than a new clinical algorithm. It combines a data audit, staged feature blocks, validation-based threshold selection, multiple model families, and explicit inspection of false-positive and false-negative errors. Prior work on multimodal learning and dementia classification motivates comparing complementary inputs and more than one evaluation metric [7], [10], [11]. The study does not infer missing CDR labels and does not claim to predict a future diagnosis.

II. Related Work

Recent systematic reviews show that ML and deep learning have been widely investigated for dementia classification using MRI and multimodal data. Borchert *et al.* reviewed AI for diagnostic and prognostic neuroimaging and highlighted the limited use of independent validation in the literature [1]. Martin *et al.* found that interpretability and robust validation remain important barriers to clinical translation [2]. Viswan *et al.* reviewed explainable AI approaches for Alzheimer's classification, while Alsubaie *et al.* surveyed deep-learning methods applied to neuroimaging [3], [4]. Kaur *et al.* compared ML and deep-learning approaches across imaging and non-imaging biomarkers [5]; Givian and Calbimonte focused on MRI-based ML for Alzheimer's disease and mild cognitive impairment [6].

Several studies examine multimodal inputs or compare model families. Golovanevsky *et al.* proposed attention-based multimodal learning, and Odusami *et al.* synthesized evidence for multimodal neuroimaging classification [7], [10]. Nan *et al.* addressed reproducible evaluation of multimodal learning [11]. Other studies investigate MRI-focused classifiers, including deep-learning methods using multimodal MRI, transfer learning, dual-attention models, and ensemble approaches [9], [13]–[16]. Research evaluating MRI-based methods in clinical data warehouses also emphasizes the difference between performance on curated research datasets and routine clinical data [12]. More recent work explicitly studies reproducibility and generalizability [17], [18].

These papers are not direct performance benchmarks for the present experiment: many use raw MRI, different cohorts,

different target definitions, or different validation schemes. Their reported accuracies should therefore not be compared numerically with this small tabular experiment. The present work instead uses the literature to motivate cautious evaluation, staged feature comparisons, and clear limits on interpretation.

Table I gives a compact thematic comparison of recent studies used to position this work. The studies are intentionally grouped by input type and methodological focus rather than ranked by headline accuracy. Their tasks differ: some classify image-level diagnostic groups, some combine modalities, and some examine interpretability or clinical generalization. This distinction matters because the current project operates on precomputed tabular measures and predicts a binary version of a contemporaneous CDR label.

III. Dataset and Preprocessing

A. Dataset audit

The supplied workbook has 436 rows and 12 attributes, including session ID, sex category, handedness, age, education, socioeconomic-status code (SES), Mini-Mental State Examination (MMSE), CDR, estimated total intracranial volume (eTIV), normalized whole-brain volume (nWBV), atlas scaling factor (ASF), and a scan-delay field. There are 416 unique participant ID prefixes across the workbook, no exact duplicate rows, and 235 distinct participants with observed CDR. CDR is missing in 201 rows, so these rows were excluded from supervised modelling rather than assigned inferred labels. Of the 235 labelled rows, 135 have CDR $= 0$ and 100 have CDR > 0; the positive group comprises 70 records with CDR 0.5, 28 with CDR 1, and two with CDR 2.

The label was encoded as $y = 0$ for CDR $= 0$ and $y = 1$ for CDR > 0. The session identifier and original CDR value were not used as predictors. Handedness was constant, and the delay field was unavailable for all labelled rows; neither was included in modelling. The analysed feature groups were: (i) demographics (age, sex, education, SES), (ii) demographics plus MRI summaries (eTIV and nWBV), and (iii) the second group plus MMSE. Nineteen labelled records lacked SES. Missing SES values were imputed using the median estimated from training records only, then applied to validation and test records. This avoids allowing held-out records to influence that preprocessing step.

B. Data partitioning

The labelled subset was divided using a stratified 60/20/20 split, with random seed 20261010. The training set contained 141 records (81 CDR-zero and 60 CDR-positive), while validation and test sets each contained 47 records (27 CDR-zero and 20 CDR-positive). The validation set was used to select decision thresholds; the test set was reserved for final metric reporting. This single split supports a controlled comparison but does not provide a stable estimate of performance variability.

The split was performed after restricting the data to rows with observed outcomes. Stratification preserved the class proportions as closely as possible in each partition. Because the supplied workbook contains repeated MRI sessions for some participants overall, a future extension should verify whether session-level records could cross partitions; the current draft reports a row-level stratified split and should not be described as participant-grouped validation unless the code explicitly groups by participant ID. This is an important reproducibility check before making broader claims.

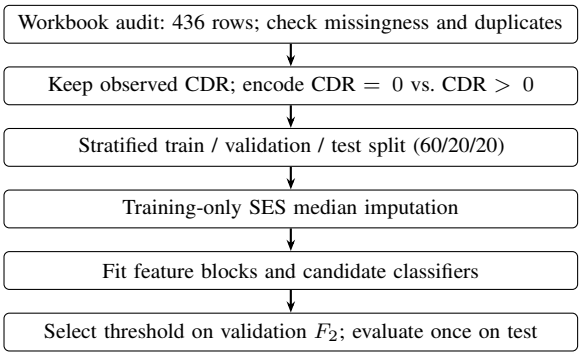


Fig. 1. Analysis pipeline used in the project.

IV. Methods

A. Analysis pipeline

Fig. 1 summarises the workflow. The analysis was implemented in R 4.3.2, with `rpart` 4.1.21 for the decision tree. Python 3.12.14 and `openpyxl` 3.1.5 were used to convert the workbook for analysis. The project scripts are described as `prepare_dataset.py` followed by `Rscript analysis.R`. The fixed random seed was 20261010.

B. Candidate models and feature blocks

Logistic regression was fitted for all three feature blocks to isolate the incremental effect of adding MRI measures and MMSE. A shallow decision tree and KNN ($k = 9$) were fitted using the full feature set. A soft-voting ensemble averaged the class-probability outputs of candidate models. A majority-class baseline was included as a simple reference. The draft project records the tree as shallow and reports the KNN setting $k = 9$; the exact tree complexity controls and any additional model-specific settings should be taken directly from the final analysis script when packaging the GitHub repository. No unreported hyperparameter values are inferred here.

For logistic regression, the estimated probability of the positive class is

$$\hat{p}(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})]}. \tag{1}$$

A case is assigned to the positive class when $\hat{p} \geq t$, where t is the decision threshold. The threshold is not fixed at 0.5 for the candidate models: it is selected on validation data by maximising F_2. The F_β measure is

$$F_\beta = (1 + \beta^2)\frac{PR}{\beta^2 P + R}, \qquad F_2 = 5\frac{PR}{4P + R}, \tag{2}$$

where P is precision and R is recall. F_2 weights recall more strongly than precision. Thresholds selected from validation

TABLE I
SELECTED RECENT LITERATURE AND RELEVANCE TO THIS STUDY

Study	Main focus / input type	Relevance and comparison boundary
Massalimova and Varol (2021) [9]	Deep learning with multimodal MRI	Illustrates image-based multimodal modelling; inputs differ from the tabular summaries used here.
Zhu *et al.* (2021) [16]	Dual-attention model on structural MRI	Shows the use of spatial MRI information not available in this workbook.
Tanveer *et al.* (2022) [15]	Transfer-learning ensemble of deep networks	Motivates ensemble comparison, but its image architecture is not reproduced in this project.
Golovanevsky *et al.* (2022) [10]	Attention-based multimodal learning	Supports the broader motivation for combining complementary information sources.
Nan *et al.* (2024) [11]	Reproducible evaluation of multimodal learning	Reinforces the importance of transparent evaluation and reproducibility.
Martin *et al.* (2023) [2]	Review of interpretable dementia ML	Highlights interpretability and clinical translation concerns.
Borchert *et al.* (2023) [1]	Systematic review of diagnostic/prognostic neuroimaging AI	Reports limited independent validation across the literature.
Bottani *et al.* (2023) [12]	MRI-based ML evaluated in clinical data	Emphasizes that curated research performance may not transfer to routine settings.
Rashid *et al.* (2023) [13]	Lightweight MRI similarity-learning network	Represents image-focused deep-learning work rather than tabular feature modelling.
Odusami *et al.* (2024) [7]	Systematic review/meta-analysis of multimodal neuroimaging	Shows the diversity of tasks and modalities in multimodal studies.
Givian and Calbimonte (2024) [6]	Review of ML using brain MRI	Provides background on MRI-based classification approaches.
Aghdam *et al.* (2025) [17]	Reproducibility and generalizability evaluation	Directly supports cautious interpretation of single-split estimates.

TABLE II
WORKBOOK AUDIT AND LABEL DISTRIBUTION

Item	Count
Source rows	436
Unique participant ID prefixes	416
Rows with known CDR	235
CDR $= 0$	135
CDR > 0	100
Missing CDR (excluded)	201
Exact duplicate rows	0
Known-target rows missing SES	19
Known-target rows missing MMSE	0

TABLE III
STRATIFIED PARTITION COUNTS

Partition	CDR $= 0$	CDR > 0	Total
Training	81	60	141
Validation	27	20	47
Test	27	20	47
Total	135	100	235

data were 0.23 for demographic logistic regression, 0.12 for demographic-plus-MRI logistic regression, 0.10 for logistic regression with MMSE, 0.12 for the decision tree, 0.33 for KNN, and 0.17 for soft voting. The baseline used 0.50.

C. Evaluation metrics

ROC-AUC measures ranking discrimination over thresholds, while accuracy measures the fraction of correct classifications at the selected threshold. Recall/sensitivity, specificity, precision, and F_2 are defined from true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$\text{Recall} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}, \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

The metrics are reported on the held-out test set of 47 records. Since thresholds were selected before test evaluation, the test results do not directly determine the chosen operating thresholds. Nevertheless, a single small test set can yield high variance and should not be interpreted as external validation.

ROC-AUC can be understood as the probability that a randomly chosen positive record receives a higher model score than a randomly chosen negative record, subject to the usual treatment of ties. It summarizes ranking ability and does not select a single operating point. In contrast, accuracy, recall, specificity, and F_2 depend on the selected threshold. This distinction is relevant here: the ensemble's AUC is marginally higher than the tree's, while the tree has the higher F_2. A metric should therefore be chosen based on a stated objective, not because it happens to be the largest number in the table. The present project does not define a clinical operating objective, so model rankings should remain descriptive.

V. RESULTS

A. Model comparison

Table IV reports the held-out test results. The soft-voting ensemble obtained the highest ROC-AUC (0.893), closely followed by the decision tree (0.892). The tree had the highest F_2 (0.893) and accuracy of 0.745. KNN achieved the highest accuracy (0.766) and specificity (0.741), but lower recall (0.800). No candidate model led on every metric, so the preferred model would depend on the relative cost of missed positive records and false alarms.

Adding MRI summaries to the demographic logistic model increased AUC from 0.719 to 0.754 and F_2 from 0.763 to 0.812. Adding MMSE raised AUC to 0.889 and F_2 to 0.885. This is an association within the present cross-sectional sample, not evidence that MMSE predicts future decline. MMSE is a cognitive measure collected alongside the CDR label and may overlap with the construct being classified.

TABLE IV
HELD-OUT TEST RESULTS AT VALIDATION-SELECTED THRESHOLDS

Model	Threshold	ROC-AUC	Accuracy	Recall	Specificity	F_2
Majority-class baseline	0.50	0.500	0.574	0.000	1.000	0.000
Logistic regression: demographics	0.23	0.719	0.532	0.900	0.259	0.763
Logistic regression: demographics + MRI	0.12	0.754	0.596	0.950	0.333	0.812
Logistic regression: demographics + MRI + MMSE	0.10	0.889	0.723	1.000	0.519	0.885
Decision tree: full features	0.12	0.892	0.745	1.000	0.556	0.893
KNN: full features ($k = 9$)	0.33	0.861	0.766	0.800	0.741	0.777
Soft-voting ensemble	0.17	0.893	0.723	0.900	0.593	0.826

B. Ensemble error analysis

At the selected ensemble threshold of 0.17, the confusion matrix contained 16 TN, 11 FP, 2 FN, and 18 TP. Thus, the ensemble detected 18 of the 20 CDR-positive test records (recall $= 0.900$) and correctly identified 16 of 27 CDR-zero records (specificity $= 0.593$). Its F_2 was 0.826. This operating point favours detection of positive records but produces a substantial number of false positives. The tree detected all 20 positive test records at its selected threshold, but produced 12 false positives; KNN produced fewer false positives (7) but missed 4 positive records. These error patterns explain why accuracy or AUC alone is insufficient for model selection.

C. Descriptive group patterns

Among records with known CDR, median age was 71 years in the CDR-zero group and 77 years in the CDR-above-zero group. Median MMSE was 29 versus 26, and median nWBV was 0.773 versus 0.723. Median eTIV was 1,415 versus 1,464 in the workbook's reported units. These are descriptive sample summaries, not causal effects. The distributions overlap, and the group differences cannot establish that any single variable causes or independently explains the recorded status.

VI. DISCUSSION

The staged feature-block comparison suggests that MRI summary variables added modest discrimination beyond demographics in this split, while adding MMSE was associated with a much larger increase. The result is plausible for a contemporaneous classification task, but it must be interpreted carefully because MMSE and CDR both reflect cognitive status. A model that includes MMSE may be useful for describing the recorded groups without being appropriate for predicting future dementia or supporting a clinical decision.

The model comparison also illustrates the dependence of performance on the chosen metric. The ensemble led on AUC, the tree led on F_2, and KNN led on accuracy and specificity. If false negatives are considered particularly costly, a high-recall operating point may be preferred; however, increased sensitivity can come with more false positives. Such choices require a defined use case and clinical input, neither of which is established in this classroom project. The small test set means that a few changed predictions would materially alter the reported values.

The majority-class baseline is also informative. It achieved 0.574 accuracy by predicting only the negative class, yet its recall and F_2 were zero. This illustrates why accuracy alone can be misleading when the target groups are not equally represented and the objective includes detecting positive cases. Conversely, the high recall of the tree and MMSE-augmented logistic model does not mean those models are automatically preferable: their specificities are 0.556 and 0.519, respectively, and their false-positive counts remain substantial. The appropriate balance would depend on the downstream context, the consequences of errors, and validation on a larger representative sample.

The descriptive medians also need restrained interpretation. Lower median MMSE and lower median nWBV in the CDR-above-zero group are consistent with the fact that these variables reflect cognitive performance and brain structure, but a difference in medians does not show that the variables are independently predictive or causal. Likewise, the eTIV medians should not be interpreted as a disease effect without accounting for factors such as age, sex, body size, and measurement conventions. A future analysis could add confidence intervals and distributional plots, but it should not turn these simple summaries into clinical claims.

VII. LIMITATIONS AND FUTURE WORK

First, CDR is missing in 201 of the 436 source rows. Excluding these records is necessary for supervised evaluation without invented targets, but the labelled subset may differ systematically from the unlabelled subset. Second, only one stratified split is reported, and the test set contains 47 records. Repeated stratified cross-validation or nested resampling should be used to estimate variability. Third, the workbook contains summary variables rather than raw MRI scans; results cannot be compared directly with image-based deep-learning studies. Fourth, the model includes MMSE, a contemporaneous cognitive measure, so the experiment should not be described as future-risk prediction. Fifth, external validation, calibration analysis, and subgroup evaluation were not performed.

Future work should repeat evaluation across multiple splits, report confidence intervals, assess calibration, document all hyperparameters, and compare performance on an independent approved cohort. The final repository should include the preprocessing script, analysis script, environment/package versions, data dictionary, and generated result tables. Sensitive data and restricted workbook content should not be uploaded publicly without checking dataset permissions.

There are also three forms of validity to consider. Internal validity is affected by the single split and by whether every pre-processing step was learned from training data only. Construct validity is affected by defining the target as CDR $= 0$ versus CDR > 0 and by including MMSE, which is related to the same broad cognitive construct as CDR. External validity is limited because the analysis uses a small labelled subset from one supplied workbook and has not been tested on a separate institution or cohort. These issues do not invalidate the exercise; they define what the results can reasonably support. A stronger follow-up would pre-register the target and evaluation protocol, group repeated sessions by participant where appropriate, run repeated or nested validation, and reserve an independent cohort for the final evaluation.

VIII. Conclusion

This project compared conventional ML classifiers for distinguishing recorded CDR-zero and CDR-above-zero groups in a supplied OASIS-1 tabular workbook. The soft-voting ensemble achieved the highest test AUC (0.893), the decision tree achieved the highest F_2 (0.893), and KNN achieved the highest accuracy (0.766) and specificity (0.741). MRI measures modestly improved the demographic logistic model in this split, while adding MMSE substantially improved the measured discrimination. Because the dataset is cross-sectional, the test sample is small, and many rows lack CDR, the findings are exploratory and do not establish clinical validity or prediction of future dementia.

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